

ibm3081_system_check_sop_v1json

```

{
  "id": "SOP-IBM3081-001",
  "title": "IBM 3081 Processor Complex: System Check & Error Logging",
  "system": "IBM 3081",
  "component": "3081 Central Processing Unit (CPU) / Service Processor",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM Maintenance Library (GA23-0061)",
  "description": "Diagnostic procedure for analyzing system checks and service processor error
logs for the IBM 3081 dyadic processor complex.",
  "steps": [
    {
      "step": 1,
      "action": "Service Processor (SVP) Console Status",
      "details": "Access the SVP console. Check the error log for 'Hard Machine Check' (HMC)
or 'Soft Machine Check' (SMC) entries.",
      "indicator": "HMC flags indicate a catastrophic failure in one of the dyadic
processors."
    },
    {
      "step": 2,
      "action": "Instruction Address Register (IAR) Extraction",
      "details": "Use the SVP to display the current IAR for both Processor A and Processor
B.",
      "check": "Cross-reference the IAR address with the microprogram map to identify the
failing instruction sequence."
    },
    {
      "step": 3,
      "action": "Check Bit Verification",
      "details": "Examine the status of the 'Machine Check' and 'Channel Check' bits in the
System Status Word.",
      "logic": "If Channel Check is active, isolate the I/O interface drivers and the
associated channel logic board."
    },
    {
      "step": 4,
      "action": "Fault Location Test (FLT) Loading",
      "details": "Initiate FLT loading from the system diskette drive via the SVP console.",
      "remedy": "Isolate the failure to a specific Thermal Conduction Module (TCM) or
Monolithic System Technology (MST) logic board."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "SVP fails to boot",
      "next": "Check SVP power supply and console cable interface."
    },
    {
      "condition": "Dyadic Processor Mismatch",
      "next": "Run CPU Logic FLTs to isolate failing TCM."
    },
    {
      "condition": "Intermittent check bit",
      "next": "Check TCM cooling and power supply voltage margins."
    }
  ]
}

```

